

Linearna regresija

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Nadgledano učenje

Oblik m. učenja kod koga program koji uči dobija skup opservacija opisanih:

- ulaznim podacima $(X_1, X_2, \dots, X_n)$ i
- željenim / tačnim izlaznim vrednostima, tako da je za svaku opservaciju poznata izlazna vrednost y_i

Zadatak programa je da “nauči” kako da novoj opservaciji dodeli tačnu izlaznu vrednost

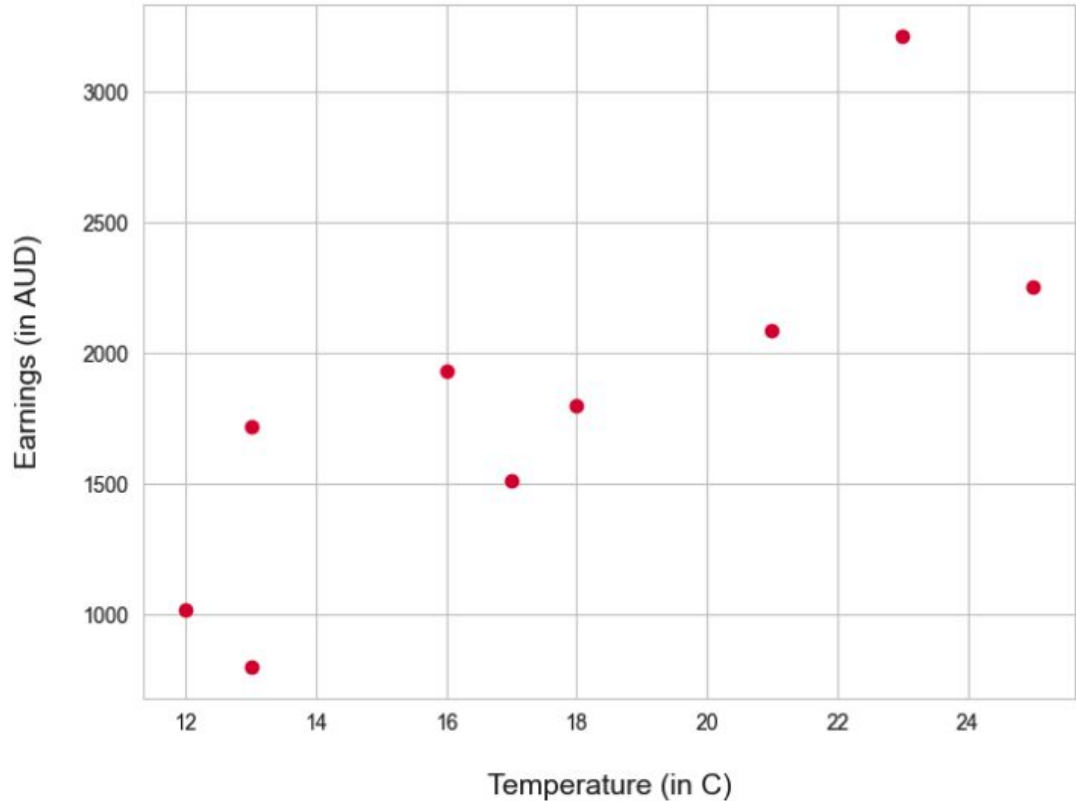
Nadgledano učenje

Izlazna vrednost vrednost određuje tip nadgledanog učenja:

- Labela: reč je o *klasifikaciji*
- Realan broj: reč je o *regresiji*

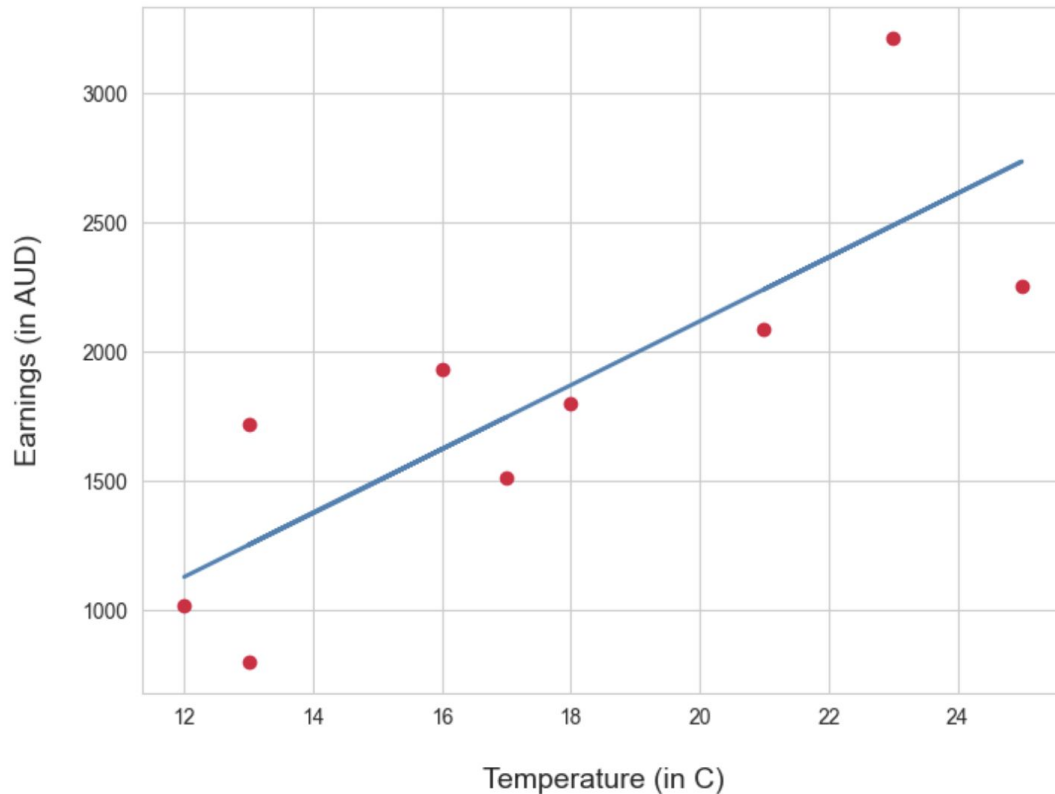
Prosta linearna regresija

Day	Takings	Temp (°C)
3-Jun	\$3,213	23
10-Jun	\$2,089	21
17-Jun	\$2,253	25
24-Jun	\$1,801	18
1-Jul	\$801	13
8-Jul	\$1,934	16
15-Jul	\$1,720	13
22-Jul	\$1,514	17
29-Jul	\$1,017	12



Prosta linearna regresija

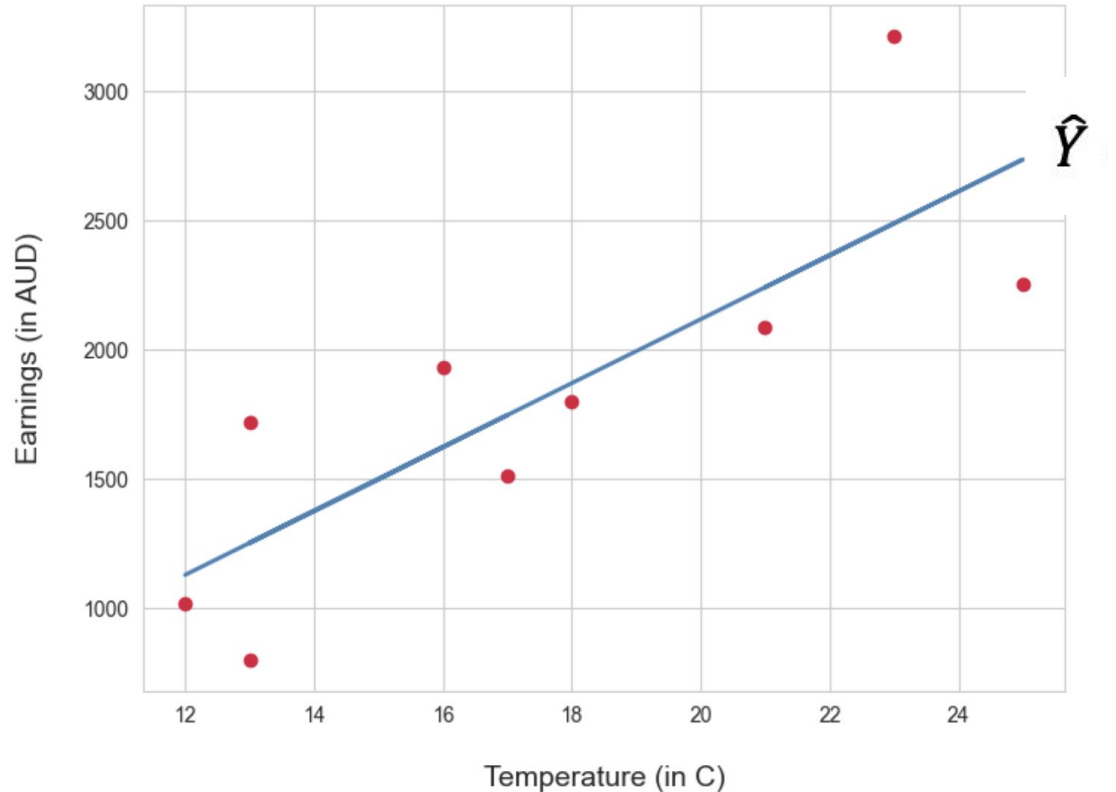
$$Y = \beta_0 + \beta_1 X + \varepsilon$$



Prosta linearna regresija

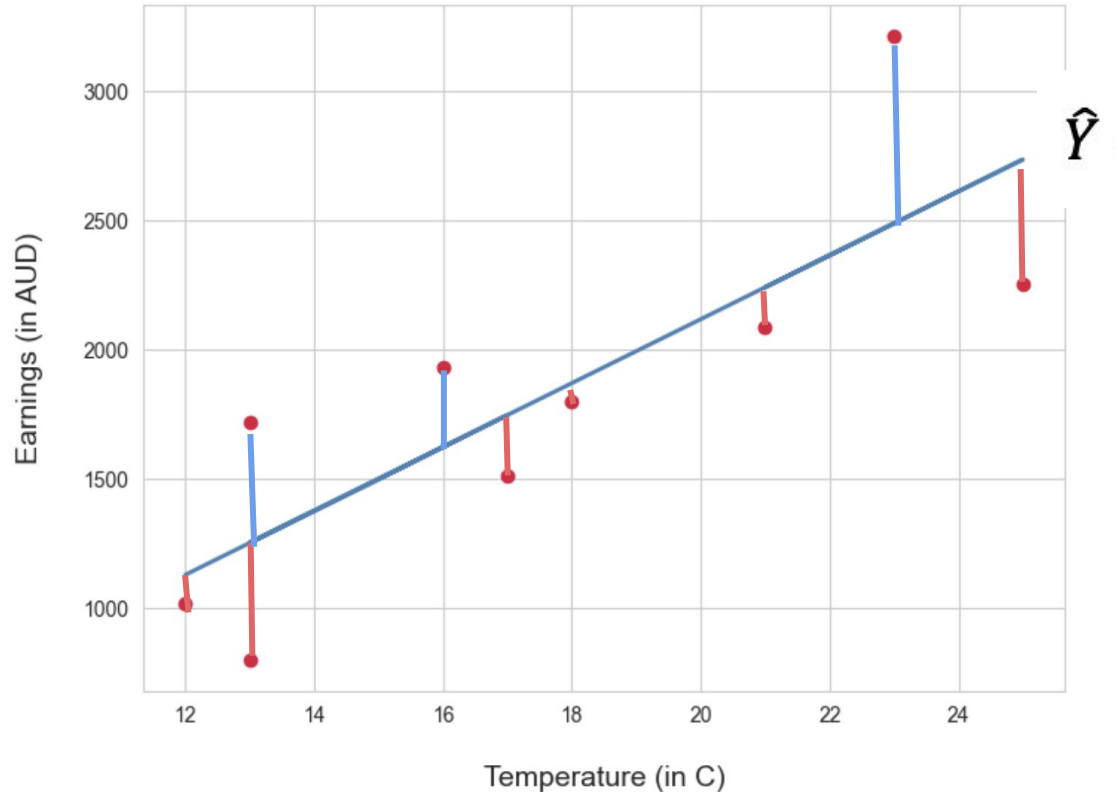
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$



Prosta linearna regresija

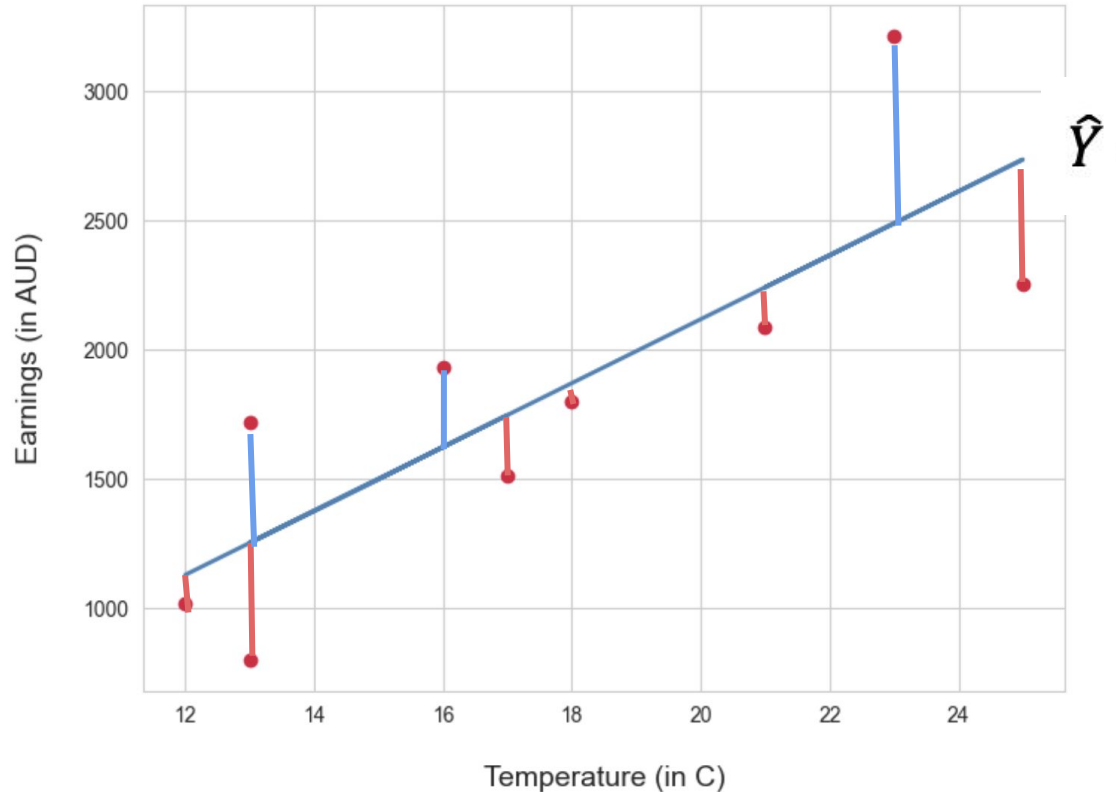
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Prosta linearna regresija

$$RSS = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

$$RSS = \sum_{i=1}^n e_i^2$$



Višestruka linearna regresija

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p.$$

Višestruka linearna regresija

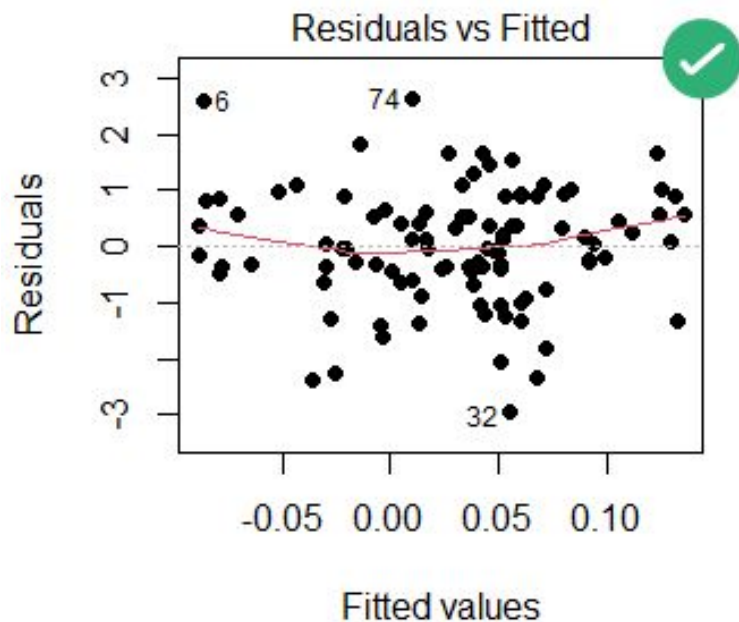
$$\begin{aligned}\text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2} - \cdots - \hat{\beta}_p x_{ip})^2.\end{aligned}$$

Primer: Calories in Starbuck Drinks

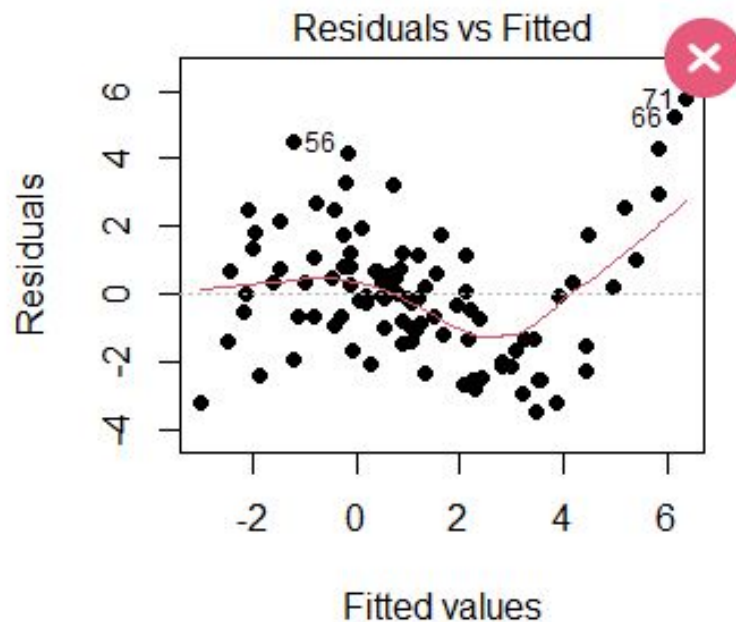
Provera pretpostavki linearnog modela

Pretpostavka linearnosti

A: Linearity assumption satisfied:

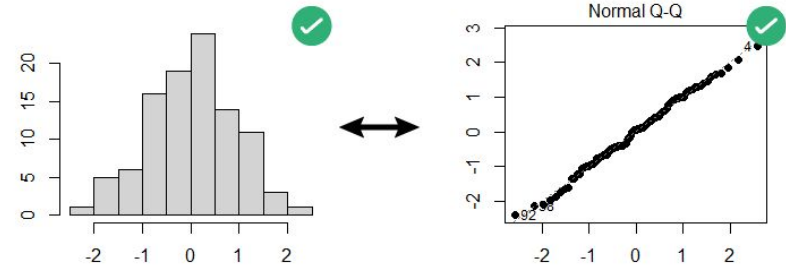


B: Linearity assumption violated:

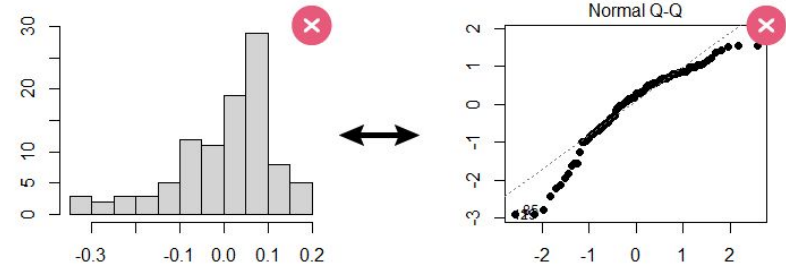


Pretpostavka normalne raspodele reziduala

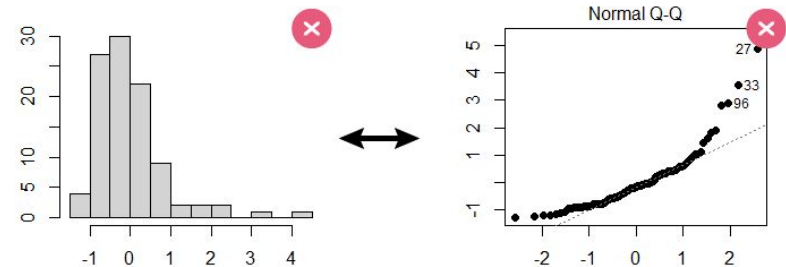
A: Normal distribution



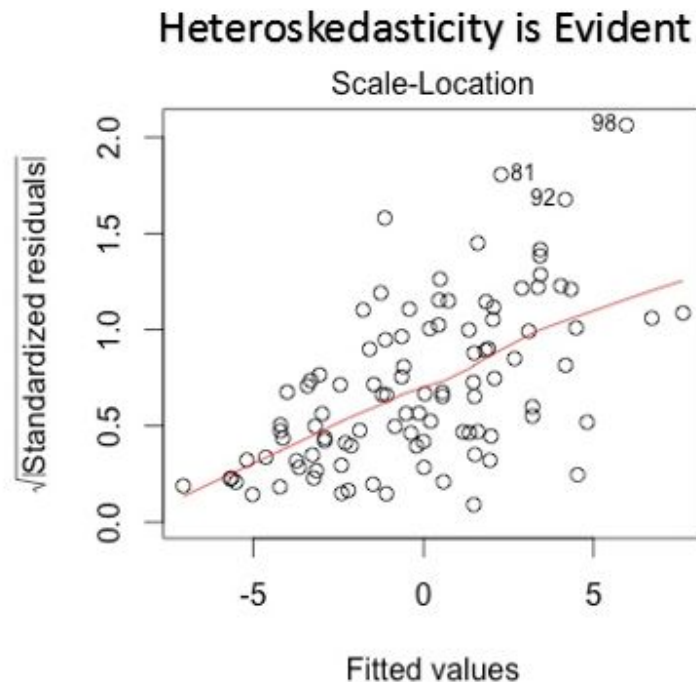
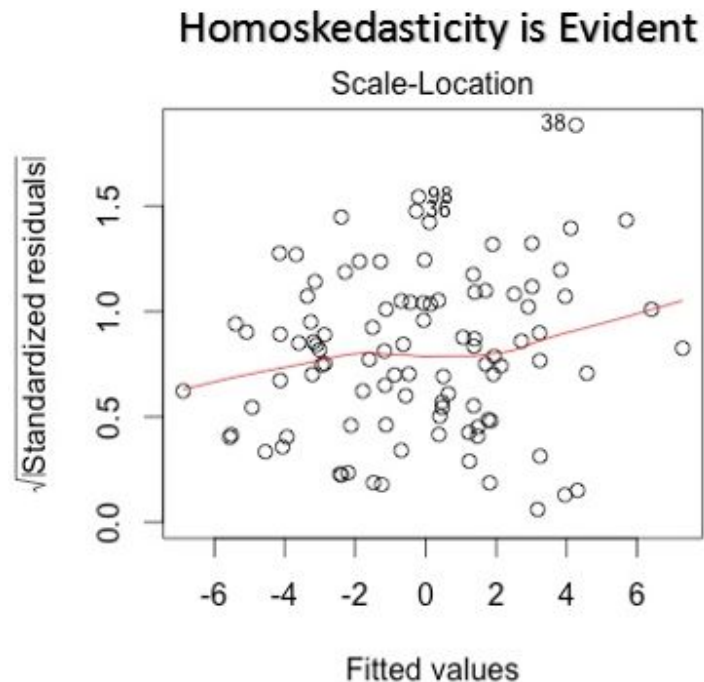
B: Left-skewed distribution



C: Right-skewed distribution

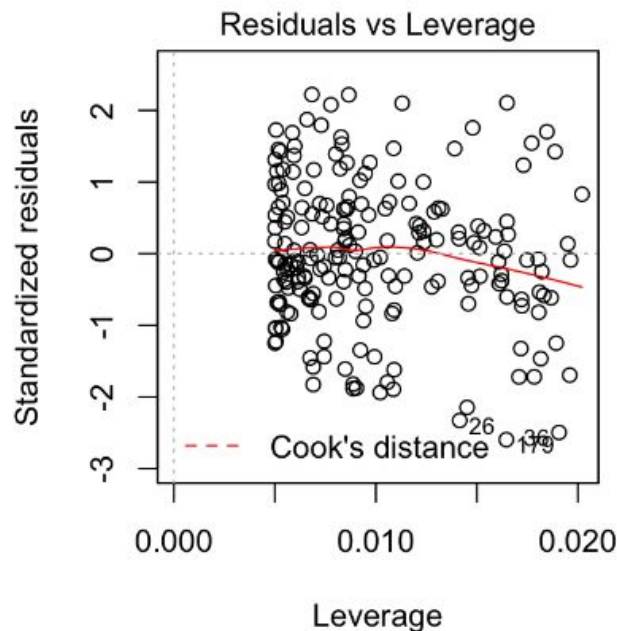


Pretpostavka ujednačene varijacije reziduala (homoskedastičnost)

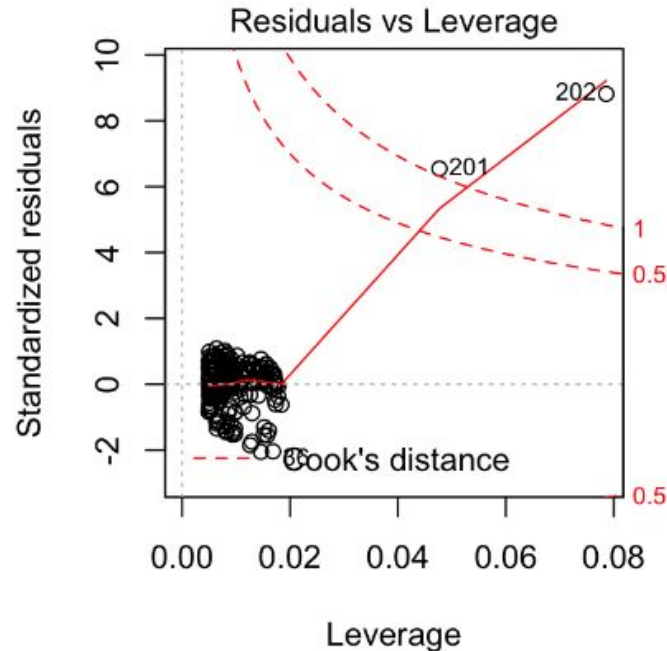


Prepostavka odsustva uticajnih opservacija

No influential points



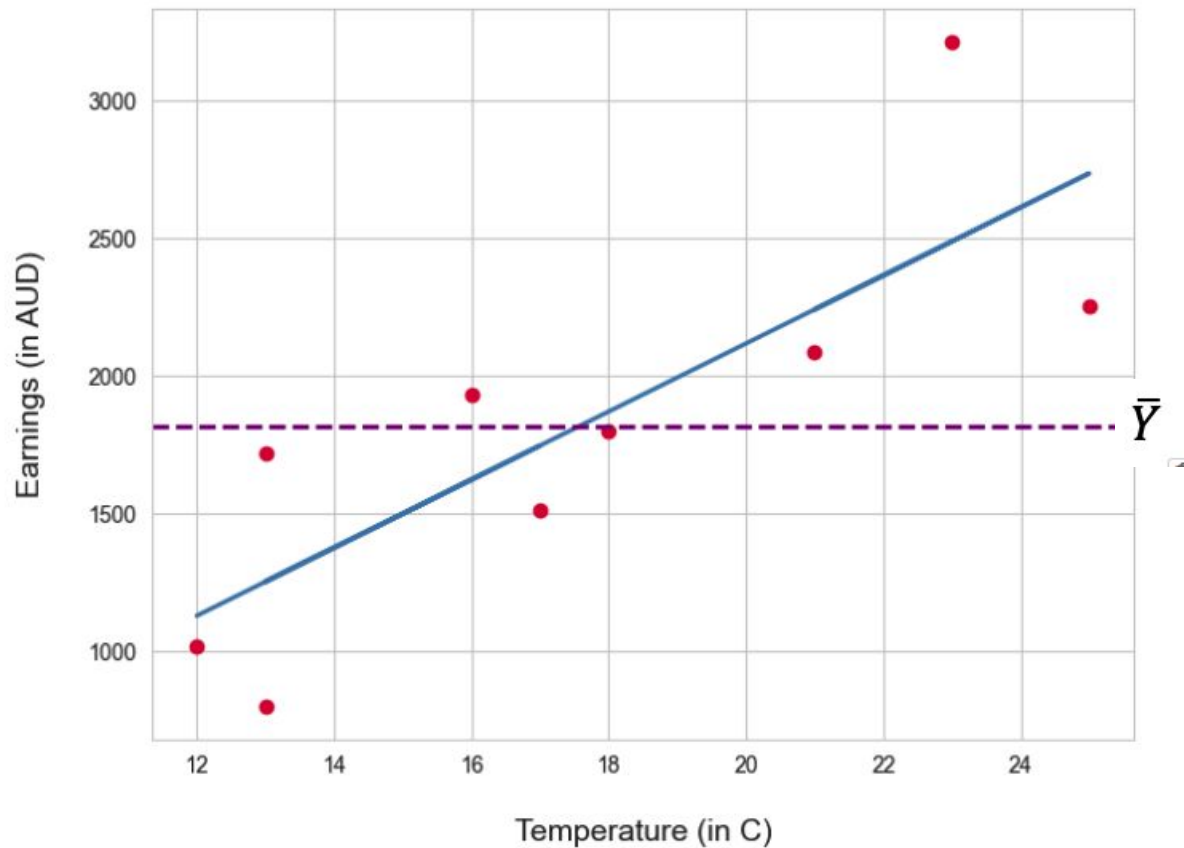
With influential points



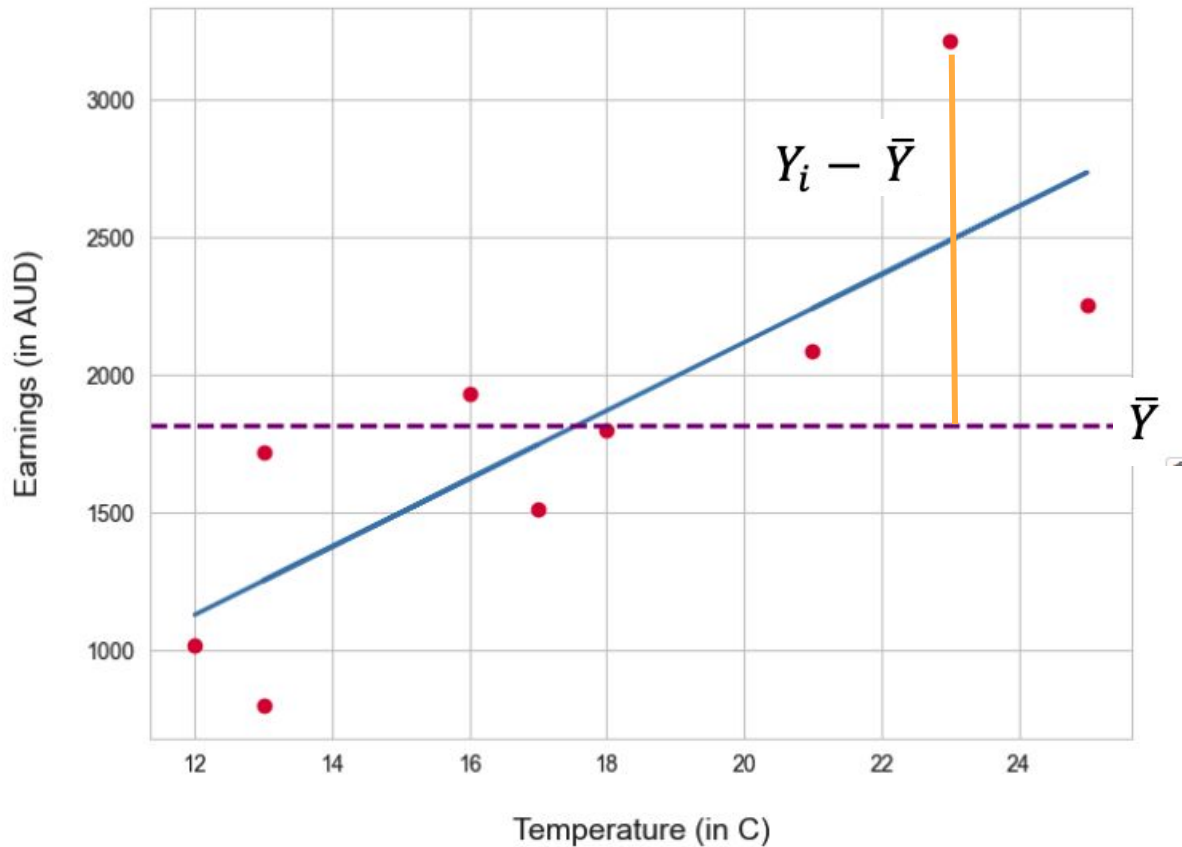
Evaluacija modela

(R^2 , RMSE)

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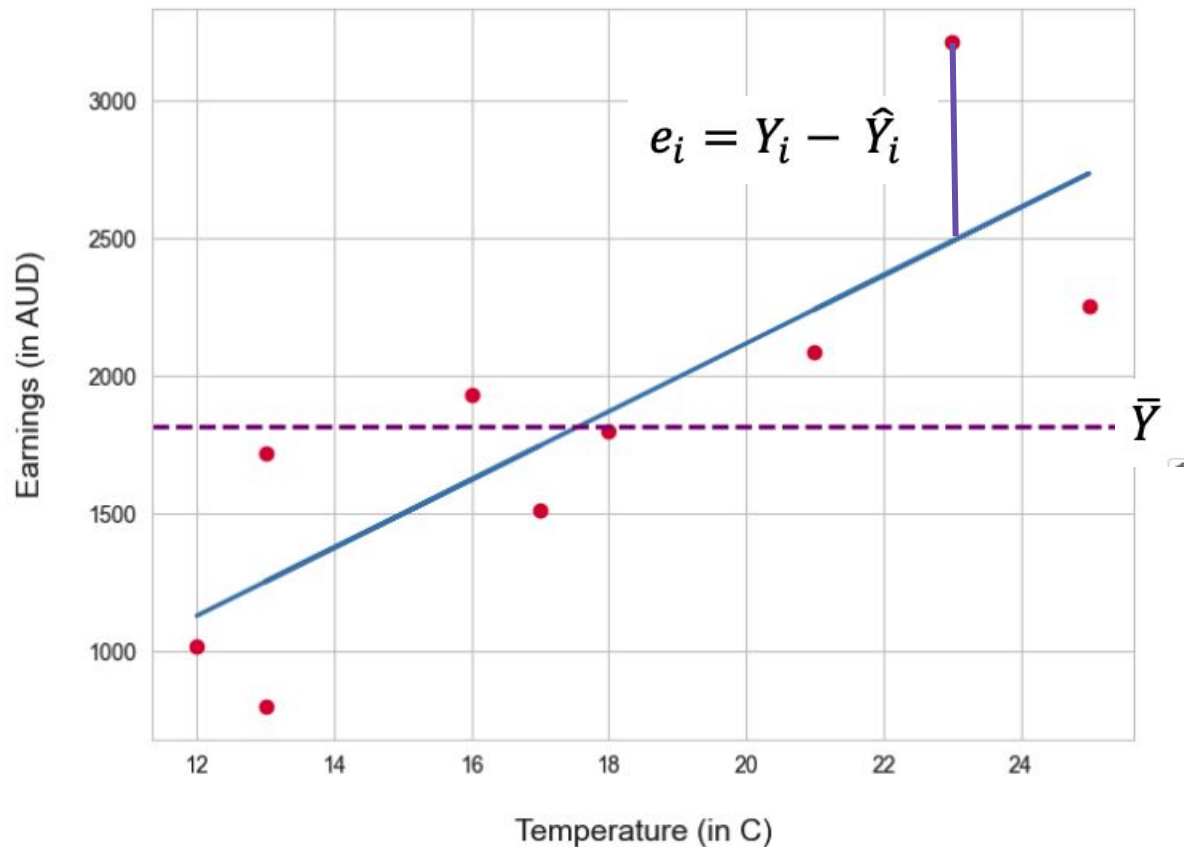


$$TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$$



TSS = Total Sum of Squares

$$RSS = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n e_i^2$$

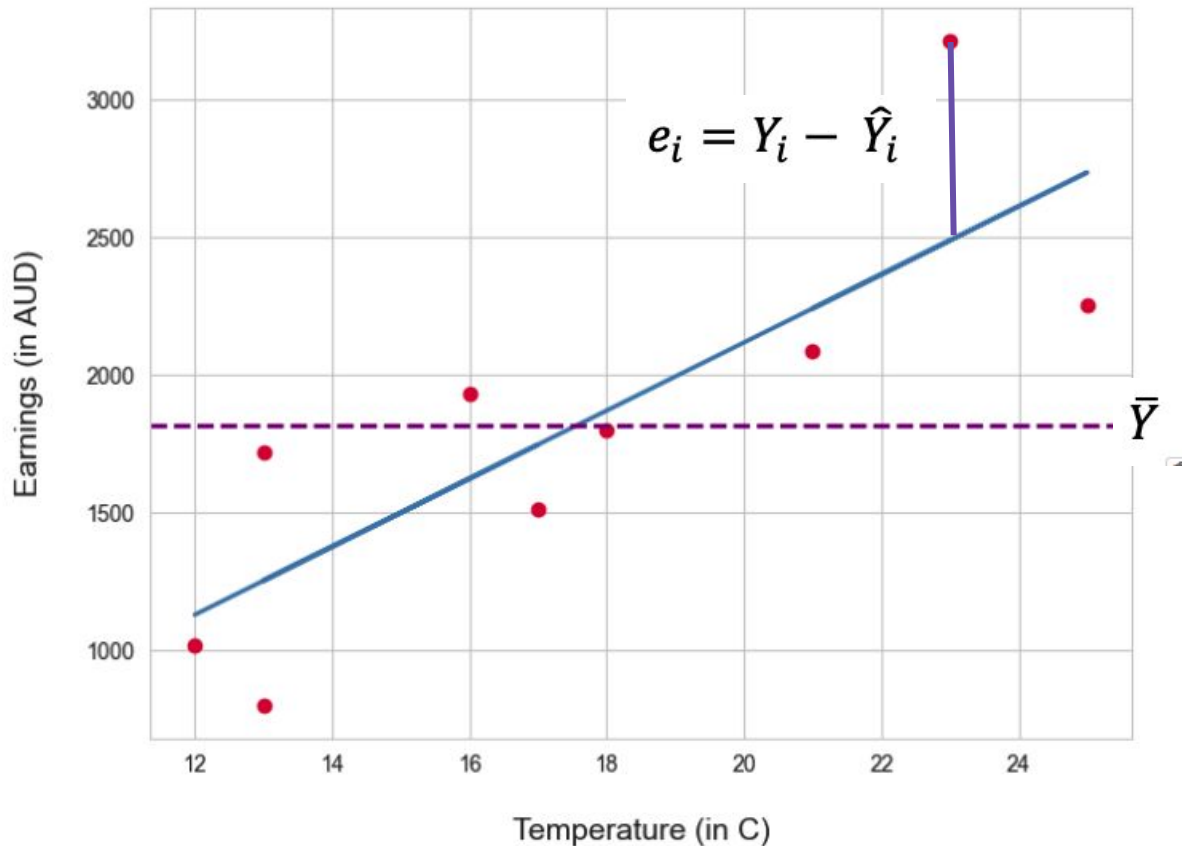


RSS = Residual Sum of Squares

$$TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

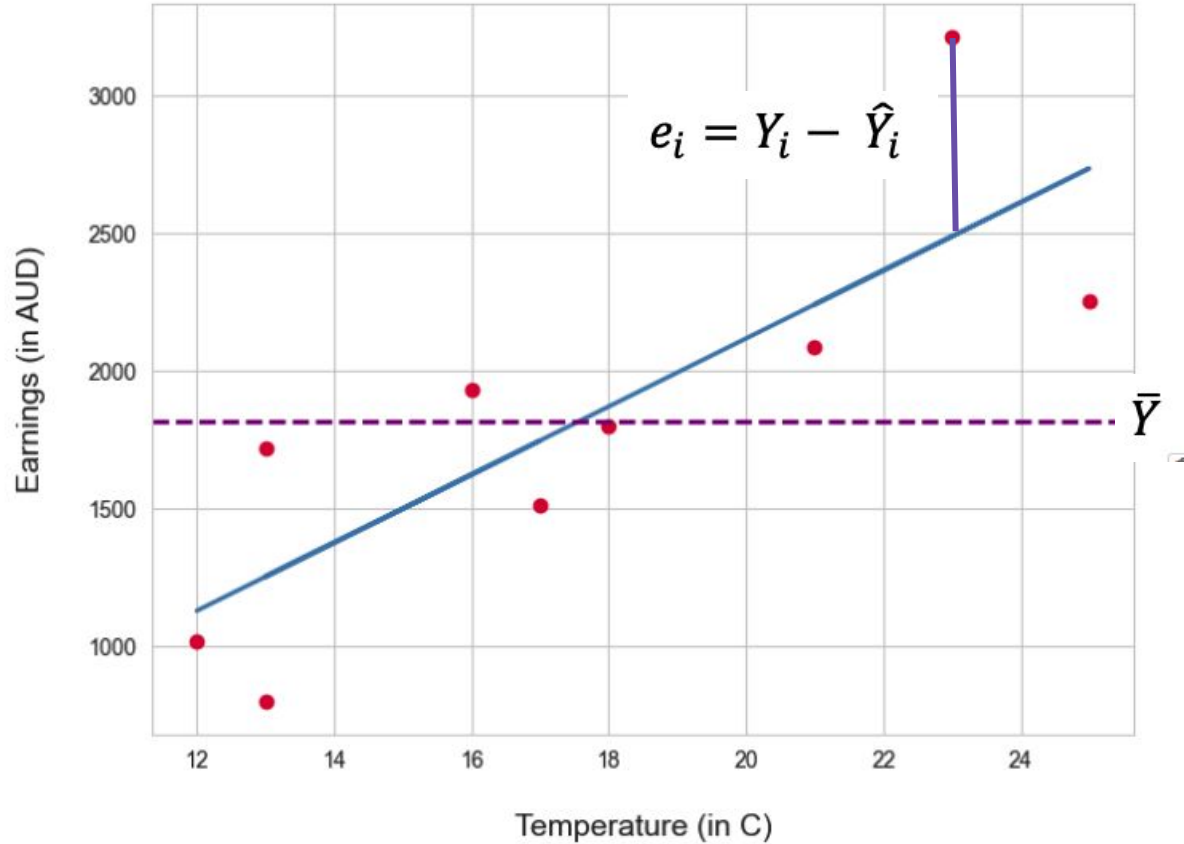
$$RSS = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n e_i^2$$

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$



$$RSS = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n e_i^2$$

$$RMSE = \sqrt{\frac{RSS}{n}}$$



RMSE = Root Mean Squared Error